Computational Game Theory **Midterm Test 01B** 3 November 2026

Name: _____

Question	1	2	3	4	5	*Total*
Points	6	6	6	6	6	30
Score						

1. **Food trucks**

   Two food trucks simultaneously choose *Station* (12 customers) or *Campus* (8 customers). A truck alone at a spot serves all its customers; two trucks at the same spot split them equally. Payoff = customers served.

   (a) (2 pts) Write the game in normal form.

   (b) (1 pts) Now, whenever the trucks share a spot, Truck 1 gets $\frac{3}{4}$ of the customers and Truck 2 gets $\frac{1}{4}$. Write the new normal form.

   (c) (3 pts) In the new game, find all pure Nash equilibria and all Pareto optimal outcomes. Is every Pareto optimal outcome an equilibrium?

Solution:

(a) Truck 1 chooses rows, Truck 2 columns.

	Campus	Station
Campus	$4, 4$	$8, 12$
Station	$12, 8$	$6, 6$

(b) When the trucks share, Truck 1 serves $\frac{3}{4} \cdot 8 = 6$ or $\frac{3}{4} \cdot 12 = 9$ customers:

	Campus	Station
Campus	$6, 2$	$8, 12$
Station	$12, 8$	$9, 3$

(c) For Truck 1, Station strictly dominates Campus ($12 > 6$, $9 > 8$). Truck 2's best response to Station is Campus ($8 > 3$). The unique pure Nash equilibrium is (Station, Campus) with payoffs $(12, 8)$.

Pareto optimal outcomes: (Campus, Station) = $(8, 12)$ and (Station, Campus) = $(12, 8)$; (Station, Station) = $(9, 3)$ and (Campus, Campus) = $(6, 2)$ are Pareto dominated by $(12, 8)$. So (Campus, Station) is Pareto optimal but not an equilibrium. In the original game both of these outcomes were equilibria; Truck 1's popularity breaks the symmetry in its favour.

2. **Iterated dominance**

	L	C	R
U	$3, 0$	$1, -1$	$0, 2$
M	$2, 1$	$0, 5$	$-1, 1$
D	$0, 1$	$2, -1$	$1, 0$

(a) (2 pts) Apply iterated elimination of strictly dominated strategies. In which order are strategies removed?

(b) (2 pts) Find the mixed Nash equilibrium of the remaining 2×2 game using indifference conditions, and both players' payoffs.

(c) (1 pts) Why can C not be removed in the first step?

(d) (1 pts) Is the equilibrium from (b) the only Nash equilibrium of the 3×3 game? Justify without computing.

Solution:

(a) M is strictly dominated by U ($3 > 2$, $1 > 0$, $0 > -1$). After removing M, column C is strictly dominated by L ($0 > -1$, $1 > -1$) and by R ($2 > -1$, $0 > -1$). The remaining game

	L	R
U	$3, 0$	$0, 2$
D	$0, 1$	$1, 0$

has no dominated strategies and no pure equilibrium (best responses cycle).

(b) Let $x = P(U)$, $y = P(L)$. Row indifference: $3y = 1 - y$, so $y = \frac{1}{4}$. Column indifference: $1 - x = 2x$, so $x = \frac{1}{3}$. Equilibrium: $(\frac{1}{3}U + \frac{2}{3}D,\ \frac{1}{4}L + \frac{3}{4}R)$, payoffs $\frac{3}{4}$ for Row and $\frac{2}{3}$ for Column. (Check in the full game: M gives $-\frac{1}{4} < \frac{3}{4}$, C gives $-1 < \frac{2}{3}$.)

(c) In the original game C is never dominated, because it is the best response to M: against M it gives 5, more than L (1) and R (1), so neither another column nor any mixture of columns is better against every row. C becomes dominated only after M has been removed.

(d) Yes. A strictly dominated strategy is never played with positive probability in a Nash equilibrium, so no equilibrium uses M; in any equilibrium Row's strategy puts no weight on M, and then C is strictly worse for Column than R, so no equilibrium uses C either. Hence every Nash equilibrium of the 3×3 game is a Nash equilibrium of the remaining 2×2 game. That game has no pure equilibrium (best responses cycle) and the indifference conditions have exactly one solution, so its only equilibrium is the one from (b). Conversely, (b) checked that this profile is an equilibrium of the full game.

3. **Market share**

A company advertises in the North (N), countrywide (C) or in the South (S); a competitor simultaneously defends the North (d_N) or the South (d_S). Entries: market-share points the company takes from the competitor (zero-sum).

	d_N	d_S
N	2	8
C	4	4
S	8	2

(a) (1 pts) Find the maximin and minimax values in pure strategies. Is there a saddle point in pure strategies?

(b) (3 pts) Write the linear program for the company's optimal strategy. What do the variable v and each constraint mean?

(c) (2 pts) The company's optimal strategy is $(\frac{1}{2}, 0, \frac{1}{2})$. Show that it guarantees 5 points, and explain why C is never used, although it is the pure maximin strategy and no single row dominates it.

Solution:

(a) Row minima $2, 4, 2$: pure maximin value 4 (row C). Column maxima $8, 8$: pure minimax value 8. Since $4 < 8$, there is no saddle point.

(b) Company, $x = (x_N, x_C, x_S)$:

$$\begin{aligned} &\text{maximize } v \\ &\text{subject to } 2x_N + 4x_C + 8x_S \geq v \quad \text{(competitor plays } d_N\text{)} \\ &\qquad\qquad 8x_N + 4x_C + 2x_S \geq v \quad \text{(competitor plays } d_S\text{)} \\ &\qquad\qquad x_N + x_C + x_S = 1, \quad x \geq 0. \end{aligned}$$

x is the company's mixed strategy and v the number of points it guarantees. Each inequality says that against one pure strategy of the competitor the company wins at least v; against a mixed strategy the payoff is an average of these, so checking the pure strategies suffices. Maximising v gives the best guarantee, the maximin value.

(c) Against d_N: $\frac{1}{2} \cdot 2 + \frac{1}{2} \cdot 8 = 5$; against d_S: $\frac{1}{2} \cdot 8 + \frac{1}{2} \cdot 2 = 5$. So the company wins 5 whatever the competitor does. Row C is not dominated by N or by S, but it is strictly dominated by the mixture $\frac{1}{2}N + \frac{1}{2}S$, which gives $(5, 5)$ against $(4, 4)$. The pure maximin strategy is the best guarantee among pure strategies only; mixing guarantees more.

4. **Correlated equilibrium in Chicken**

Two drivers simultaneously choose to Yield (Y) or Dare (D):

	Y	D
Y	$6, 6$	$2, 7$
D	$7, 2$	$0, 0$

Let $\alpha, \beta, \gamma, \delta$ be the probabilities of (Y,Y), (Y,D), (D,Y), (D,D).

(a) (3 pts) Write and simplify the correlated equilibrium conditions. Give one correlated equilibrium.

(b) (3 pts) A device recommends (Y,Y) with probability $\frac{3}{5}$ and (Y,D), (D,Y) with $\frac{1}{5}$ each, and privately tells each driver only his own recommendation. What does each driver get if both obey? Is it a correlated equilibrium? If not, which recommendation is not followed, and what does a deviating driver gain?

Solution:

(a) Driver 1 told Y: $6\alpha+2\beta \geq 7\alpha+0\beta$, i.e. $2\beta \geq \alpha$. Driver 1 told D: $7\gamma+0\delta \geq 6\gamma+2\delta$, i.e. $\gamma \geq 2\delta$. Driver 2 told Y: $6\alpha+2\gamma \geq 7\alpha$, i.e. $2\gamma \geq \alpha$. Driver 2 told D: $7\beta \geq 6\beta+2\delta$, i.e. $\beta \geq 2\delta$. Together with $\alpha, \beta, \gamma, \delta \geq 0$, $\alpha+\beta+\gamma+\delta = 1$.

Examples of correlated equilibria (any one suffices): the pure Nash equilibrium (Y,D), i.e. $\beta = 1$; the "traffic light" $\beta = \gamma = \frac{1}{2}$; or $\alpha = \beta = \gamma = \frac{1}{3}$, $\delta = 0$. Each satisfies all four inequalities.

(b) If both obey, each driver gets $\frac{3}{5}\cdot 6 + \frac{1}{5}\cdot 2 + \frac{1}{5}\cdot 7 = \frac{27}{5}$. Not a correlated equilibrium: the condition $2\beta \geq \alpha$ fails ($\frac{2}{5} < \frac{3}{5}$). A driver told Y knows the other was told Y with probability $\frac{3/5}{4/5} = \frac{3}{4}$ and D with probability $\frac{1}{4}$: obeying gives $\frac{3}{4}\cdot 6 + \frac{1}{4}\cdot 2 = 5$, daring gives $\frac{3}{4}\cdot 7 + \frac{1}{4}\cdot 0 = \frac{21}{4}$. So he ignores the recommendation Y and dares, gaining $\frac{1}{4}$ whenever he is told Y. (A driver told D knows the other yields; $7 > 6$, so he obeys.)

5. **Quantal response and commitment**

	L	R
T	$2,2$	$4,0$
B	$1,0$	$3,1$

(a) (2 pts) Column plays $(\frac{1}{2}, \frac{1}{2})$. Find Row's logit response for $e^{\lambda} = 3$.

(b) (1 pts) Why does Row play the dominated B with positive probability for every finite λ, and why does (a) not depend on Column's strategy?

(c) (2 pts) Row is a leader who commits to $p = P(B)$; Column observes p and best responds. Write the linear program for each follower action and find the strong Stackelberg equilibrium.

(d) (1 pts) Compare the leader's payoff in the strong Stackelberg equilibrium with her payoff in the Nash equilibrium of the simultaneous game. Why does she commit to the strictly dominated B so often?

Solution:

(a) Against any strategy of the column player, T pays exactly 1 more than B: $U(T) = 2y + 4(1-y)$ and $U(B) = y + 3(1-y)$, so $d = U(T) - U(B) = 1$. Hence

$$P(T) = \frac{1}{1+e^{-\lambda d}} = \frac{1}{1+\frac{1}{3}} = \frac{3}{4}, \qquad P(B) = \frac{1}{4}.$$

(b) The logit response gives every action a probability proportional to $e^{\lambda \cdot \text{payoff}}$, which is positive for finite λ; a worse action is played less often, but never with probability 0. Only the payoff *difference* matters: $P(B) = \frac{1}{1+e^{\lambda d}}$. Since B is worse than T by exactly $d = 1$ against *every* column strategy, $P(B) = \frac{1}{1+e^{\lambda}}$ is the same whatever the column player does.

(c) Follower: $u_F(L) = 2(1-p)$, $u_F(R) = p$; leader: $u_L(L) = 2(1-p) + p = 2 - p$, $u_L(R) = 4(1-p) + 3p = 4 - p$.

$$\begin{aligned} LP_L &: \max\ 2-p \ \text{ s.t. } 2(1-p) \geq p,\ 0 \leq p \leq 1 \quad \Rightarrow p = 0,\ \text{value } 2, \\ LP_R &: \max\ 4-p \ \text{ s.t. } p \geq 2(1-p),\ 0 \leq p \leq 1 \quad \Rightarrow p = \tfrac{2}{3},\ \text{value } \tfrac{10}{3}. \end{aligned}$$

Strong Stackelberg equilibrium: the leader plays B with probability $\frac{2}{3}$, the follower plays R; payoffs $(\frac{10}{3}, \frac{2}{3})$.

(d) The unique Nash equilibrium is (T, L) with payoffs $(2, 2)$ (T is dominant, L is the best response). Committing to B often enough makes R a best response for the follower, and against R the leader gets 3 or 4 instead of at most 2. A dominated action is never played in Nash equilibrium, but it can be worth committing to, because it changes the follower's response.